



Template for Reporting Results of Biomarker Testing of Specimens from Patients with Gastrointestinal Stromal Tumors

Version: 1.2.0.0

Protocol Posting Date: September 2026

This biomarker template is not required for accreditation purposes but may be used to facilitate compliance with CAP Accreditation Program Requirements.

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Glossary:

Author: Expert who is a current member of the Cancer Committee, or an expert designated by the chair of the Cancer Committee.

Expert Contributors: Includes members of other CAP committees or external subject matter experts who contribute to the current version of the protocol.

Accreditation Requirements

Completion of the template is the responsibility of the laboratory performing the biomarker testing and/or providing the interpretation. When both testing and interpretation are performed elsewhere (e.g., a reference laboratory), synoptic reporting of the results by the laboratory submitting the tissue for testing is also encouraged to ensure that all information is included in the patient's medical record and thus readily available to the treating clinical team.

Summary of Changes

v 1.2.0.0

- GIST title / lineage is now inclusive of all GIST sites and not limited to stomach
- Updates to content and explanatory notes
- Addition of new section headings "Diagnostic Immunohistochemistry" and "Molecular Subtype Screening"
- Update to KIT Exons Assessed question
- Update to PDGFRA Mutational Analysis question, now multiselect
- Addition of optional NTRK Fusion Gene Analysis question

Reporting Template

Protocol Posting Date: September 2026

Select a single response unless otherwise indicated.

CASE SUMMARY: (GIST Biomarker Reporting Template)

Standard(s):

Completion of the template is the responsibility of the laboratory performing the biomarker testing and / or providing the interpretation. When both testing and interpretation are performed elsewhere (e.g., a reference laboratory), synoptic reporting of the results by the laboratory submitting the tissue for testing is also encouraged to ensure that all information is included in the patient's medical record and thus readily available to the treating clinical team.

Fixative type, time to fixation (cold ischemia time), and time of fixation should be reported if applicable in this template or in the original pathology report.

Gene names should follow recommendations of The Human Genome Organization (HUGO) Nomenclature Committee (www.genenames.org; accessed July 6, 2026).

All reported gene sequence variations should be identified following the recommendations of the Human Genome Variation Society (<https://hgvs-nomenclature.org/stable/>; accessed July 6, 2026).

GIST BIOMARKER TESTS (Note [A](#))

Immunohistochemical Studies (Note [B](#))

Diagnostic Immunohistochemistry

+KIT (CD117)

☐ Positive
☐ Negative

+DOG1 (ANO1)

☐ Positive
☐ Negative

Molecular Subtype Screening

+SDHA

☐ Intact
☐ Deficient
☐ Cannot be interpreted due to absent internal control

+SDHB

☐ Intact
☐ Deficient
☐ Cannot be interpreted due to absent internal control

Other IHC Studies

+Test Name (repeat this question as needed up to 10 times): _____

☐ Positive
☐ Negative
☐ Other (specify): _____

Molecular Genetic Studies (Note [C](#))

+KIT Mutational Analysis (Note [D](#))

- ☐ No mutation detected
☐ Mutation identified (specify): _____
☐ Cannot be determined: _____

+KIT Exons Assessed (Note [D](#)) (select all that apply)

- ☐ Exon 8
☐ Exon 9
☐ Exon 11
☐ Exon 13
☐ Exon 14
☐ Exon 17
☐ Exon 18
☐ Other (specify): _____

+PDGFRA Mutational Analysis (Note [E](#)) (select all that apply)

- ☐ No mutation detected
☐ PDGFRA p.Asp842Val (D842V) mutation identified
☐ Other PDGFRA mutation identified (specify): _____
☐ Cannot be determined: _____

+PDGFRA Exons Assessed (Note [E](#)) (select all that apply)

- ☐ Exon 12
☐ Exon 14
☐ Exon 18
☐ Other (specify): _____

+SDH Mutational Analysis (Note [G](#))

- ☐ No mutation detected
☐ Mutation identified (specify): _____
☐ Cannot be determined: _____

+NF1 Mutational Analysis (Note [H](#))

- ☐ No mutation detected
☐ Mutation identified (specify): _____
☐ Cannot be determined: _____

+BRAF Mutational Analysis (Note [I](#)) (select all that apply)

- ☐ No BRAF mutation detected
☐ BRAF V600E (c.1799T>A, exon 15) mutation identified
☐ Other BRAF mutation identified (specify): _____
☐ Cannot be determined: _____

+BRAF Fusion Gene Analysis (Note [I](#))

- ☐ No fusion detected
☐ Fusion identified (specify): _____
☐ Cannot be determined: _____

+FGFR1 Fusion Gene Analysis (Note [I](#))

- ☐ No fusion detected
☐ Fusion identified (specify): _____
☐ Cannot be determined: _____

+NTRK Fusion Gene Analysis (Note [J](#))

- ☐ No fusion detected
☐ Fusion identified (specify): _____
☐ Cannot be determined: _____

Other Gene Mutational Analysis

+Mutation Name (repeat this question as needed up to 10 times): _____

- ☐ No mutation detected
☐ Mutation identified (specify): _____
☐ Cannot be determined (explain): _____

Other Fusion Gene Analysis

+Fusion Name (repeat this question as needed up to 10 times): _____

- ☐ No fusion detected
☐ Fusion identified (specify): _____
☐ Cannot be determined (explain): _____

COMMENTS

Comment(s): _____

Explanatory Notes

A. Reporting Nomenclature

Consistent gene mutation nomenclature is essential for efficient and accurate reporting.¹ The following are examples as recommended by the Human Genome Variation Society (HGVS) for description of variant changes.² It is also preferred that protein alterations are mentioned in the report in addition to genomic coordinates.

Examples of DNA, RNA, and Protein Nomenclature

DNA: A, G, C, T (example: c.957A>T)

RNA: a, g, c, u (example: r.957 a>u)

Protein: 3-letter amino acid code, X= Stop codon (example: p. Glu78Gln)

Examples of Nomenclatures for Types of Sequence Variants

<u>Types of Variation</u>	<u>Examples</u>
Substitution	c.123A>G
Deletion	c.123delA, c.586_591delTGGTCA or c.586_591del6
Duplication	c.123dupA, c.586_591dupTGGTCA or c.586_591dup6
Insertion	c.123_124insC, c.1086_1087insGCGTGA
Frame shift	p. Arg83fs or p. Arg83SerfsTer15
Deletion/insertion "indel"	c.112_117delAGGTCAinsTG

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B. Immunohistochemical Analysis

Because small-molecule kinase inhibitor therapy is highly effective in the treatment of GIST, it has become imperative to distinguish GIST from histologic mimics, mainly leiomyoma, leiomyosarcoma, schwannoma, and desmoid fibromatosis.^{1,2} Immunohistochemistry is instrumental in the workup of GIST. For the initial workup of GIST, a basic immunohistochemical panel including CD117 (KIT), DOG1 (ANO1), desmin, and S100 protein is recommended. GIST is immunoreactive for KIT (CD117) (95%) and/or DOG1(>99%).^{3,4,5} KIT immunoreactivity is usually strong and diffuse but can be more focal in unusual cases (Figure 1, A, and B). It is not unusual for GIST to exhibit dot-like perinuclear staining (Figure 1, C), while less commonly, cases exhibit membranous staining (Figure 1, D). These patterns do not correlate with mutation type or response to therapy. Most KIT-negative/DOG1 positive GIST are gastric or extra-visceral and almost invariably harbor a *PDGFRA* mutation.⁶ DOG1 expression is not related to mutational status, and it is a useful marker to identify KIT-negative GISTs.^{4,5} Approximately 70% of GISTs are positive for CD34, 30% to 40% for smooth muscle actin, 5% for S100 protein (usually focal), 5% for desmin (usually focal), and 1% to 2% for keratin (weak/focal).⁷ PanTrk immunohistochemistry may be positive in GIST independent of *NTRK* fusion, and this immunostain is not recommended in GIST.

Succinate dehydrogenase (SDH) is a mitochondrial enzyme comprising four subunits (SDHA, SDHB, SDHC, and SDHD) that is involved in the Krebs cycle. Genetic alteration of any of the four subunits

results in SDH dysfunction and subsequent loss of SDHB expression by immunohistochemistry. SDH-deficient GISTs are characterized by early onset, gastric location, female predominance, multifocality, a plexiform/multinodular growth pattern, frequent lymph node metastases, and epithelioid morphology; they are *KIT* and *PDGFRA* wild-type.^{8,9} One-half to two-thirds of SDH-deficient GISTs are associated with a pathogenic germline mutation in an *SDH* subunit gene (*SDHx*), most commonly *SDHA*. A substantial proportion of the remaining tumors demonstrate *SDHC* promoter hypermethylation, particularly in the setting of Carney triad. Rare tumors harbor biallelic somatic *SDHx* alterations or remain molecularly unexplained despite comprehensive testing. Thus, it is recommended that all gastric GISTs be screened for loss of SDH by immunohistochemistry, best accomplished by immunostaining for SDHB, which is lost in all SDH-deficient GIST, independent of the SDH-subunit that is inactivated.^{10,11,12,13} Mutations in *SDHA* are detected in ~30% of SDH-deficient GISTs and loss of expression of SDHA specifically identifies tumors with *SDHA* mutations; other SDH-deficient GISTs show normal (intact) cytoplasmic staining.^{14,15} Patients with SDH-deficient GIST should be referred to a genetic counselor for appropriate workup.

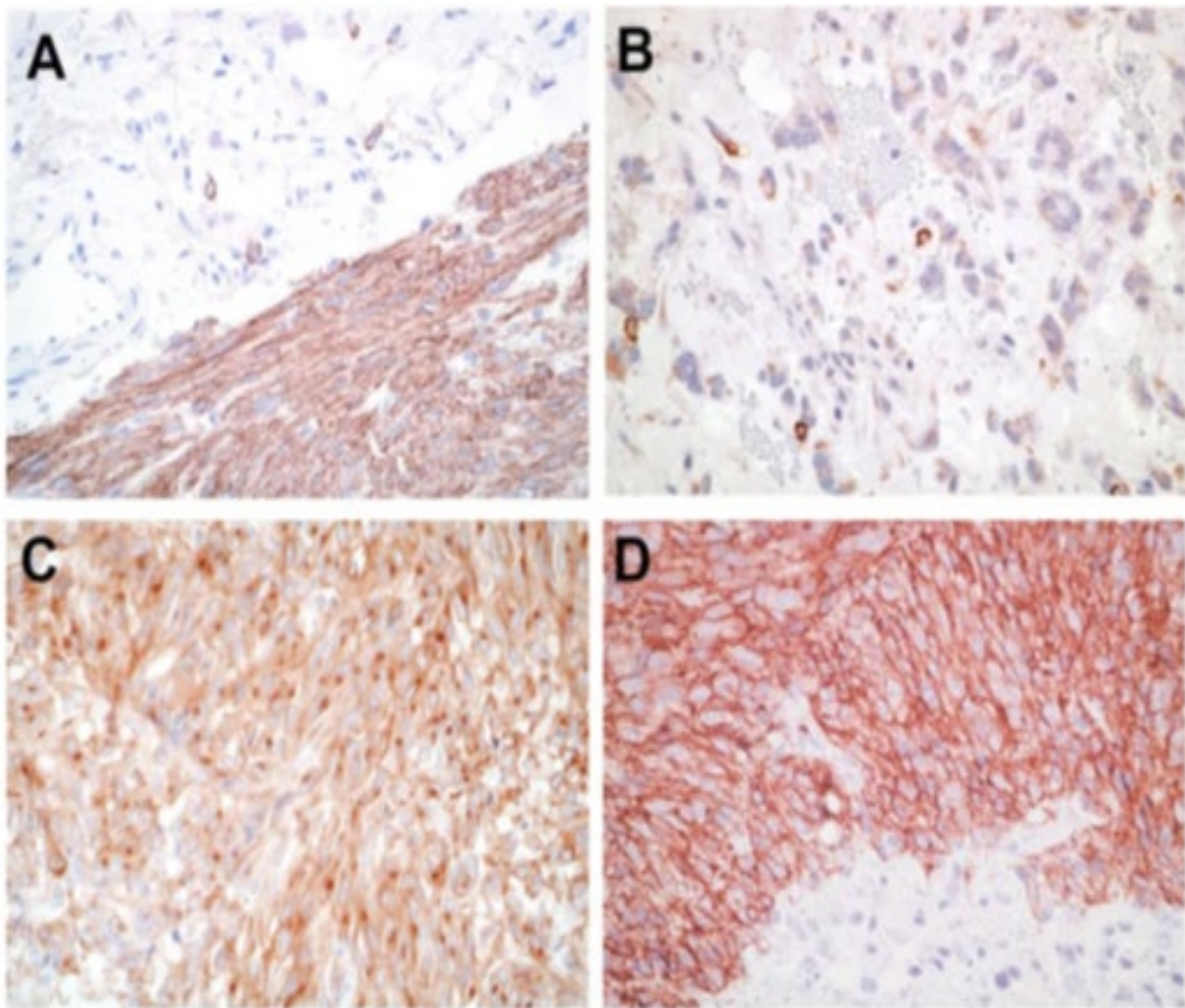


Figure 1. Patterns of KIT staining in gastrointestinal stromal tumor (GIST). A, Diffuse and strong immunoreactivity in a typical GIST. B, Focal and weak staining in an epithelioid gastric GIST with a

PDGFRA mutation. C, Dot-like perinuclear staining. D, Membranous pattern staining. (Original magnification X400)

References

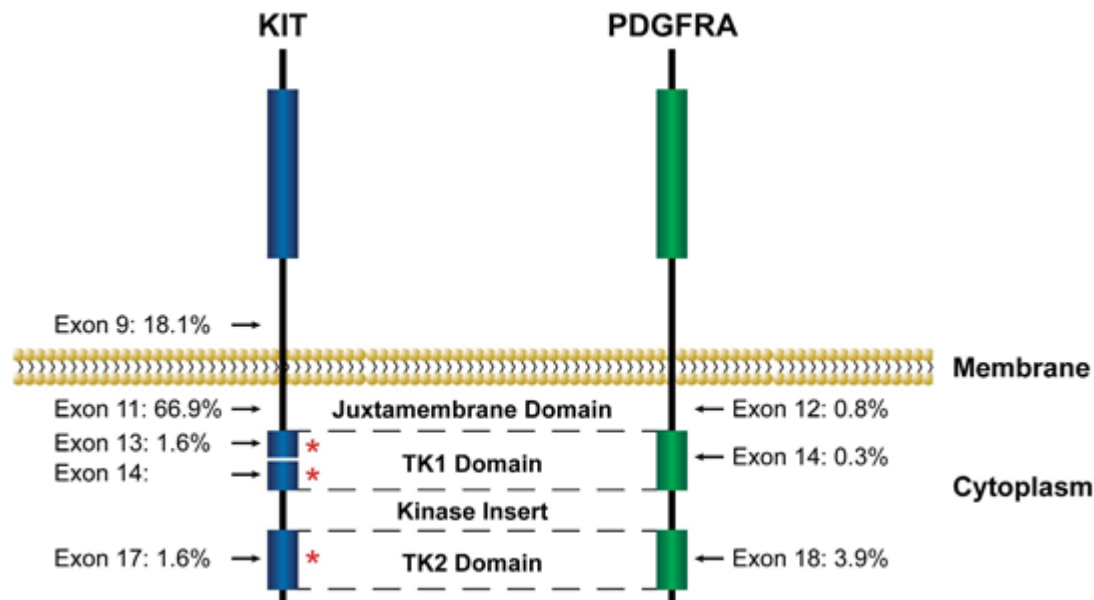
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C. Molecular Analysis

Approximately 75% of GISTs possess activating mutations in *KIT*, whereas another 10% have activating mutations in *PDGFRA*.^{1,2,3,4} The resulting mutant proteins demonstrate ligand-independent activation. *KIT* and *PDGFRA* each contain 21 exons. However, mutations cluster within “hotspots”: exons 9, 11, 13, and 17 in *KIT* and exons 12, 14, and 18 in *PDGFRA* (Figure 2). About 10 to 15% of GISTs appear to be negative for both *KIT* and *PDGFRA* mutations. The most recent NCCN Task Force on GIST strongly encouraged that *KIT* and *PDGFRA* mutational analysis be performed in unresectable and metastatic GISTs for which use of tyrosine kinase inhibitors (TKIs) is considered; mutation testing should also be considered for patients with primary disease, particularly if high-risk. *KIT* and *PDGFRA* mutation status can be readily determined from paraffin-embedded tissue. Secondary (acquired) resistance mutations can develop in the setting of long-term imatinib mesylate treatment. These are usually point mutations that occur most commonly in *KIT* exons 13, 14, 17, and 18.⁵

Recent studies focusing on the molecular classification of GIST recognized two major subgroups: succinate dehydrogenase (SDH)-competent (92-95%) and SDH-deficient GIST (5-8%), both of which can arise in the sporadic or familial setting.^{6,7} SDH-competent GISTs include tumors with mutations of *KIT* or *PDGFRA*, as well as those with mutations in *NF1* (2-6%) or *BRAF* ($\leq 1\%$) and rarely fusions involving *FGFR1* or *BRAF*.^{8,9,10,11,12,13} Other rare molecular genetic events have been described including *PIK3CA* mutation (which is not necessarily independent of *KIT*/*PDGFRA* activation), *NTRK* fusion (some of these tumors may be better characterized as non-GISTs), and *RAS* family genes (very uncommon).

SDH-deficient GISTs are characterized by genetic or epigenetic inactivation of any SDH subunit. As mentioned previously, up to two-thirds of SDH-deficient GISTs are associated with a pathogenic germline mutation in an *SDH* subunit gene (*SDHx*), most commonly *SDHA*, with a substantial proportion of the remaining tumors demonstrating *SDHC* promoter hypermethylation. Somatic biallelic inactivation of an *SDHx* gene is considered rare.^{14,15,16} The majority of pediatric GIST arise in Carney triad (SDH-deficient GIST, pulmonary chondroma, extra-adrenal paraganglioma; typically due to *SDHC* promoter methylation) or Carney-Stratakis syndrome (SDH-deficient GIST and paraganglioma due to germline mutation in an *SDHx* gene). Germline mutations in *SDHx* can also lead to SDH-deficient renal cell carcinomas and pituitary tumors.



* Refers to exons involved most frequently by secondary/acquired mutations.

Figure 2. Locations and frequency of activating *KIT* and *PDGFRA* mutations in GIST. This was adapted with permission from Heinrich et al.¹ Copyright 2003 by the American Society of Clinical Oncology. All rights reserved.

KIT and *PDGFRA* are excellent targets for small-molecule tyrosine kinase inhibitors (TKIs), and compounds of this class, including imatinib mesylate, sunitinib malate, avapritinib (specifically indicated in *PDGFRA* D842V-mutant tumors), regorafenib, and ripretinib, are Food and Drug Administration-approved for the treatment of GIST.^{17,18,19,20} SDH-deficient GISTs are usually resistant to imatinib but may respond to sunitinib.^{6,21} Clinical interest in mutation status is driven by differential response to TKIs, which may drive drug selection.²² Detection of secondary (acquired) resistance mutations may also impact drug selection as their significance is further defined.

NTRK fusions in GIST are very uncommon. 10 to 15% of tumors may stain positively for pan-TRK, reflecting either wild-type TRK protein expression or non-specific staining, and NGS or FISH are preferred to immunohistochemistry for the detection of *NTRK* fusions in GIST.^{23,24,25}

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D. KIT Mutational Analysis

The most common mutations affect the juxtamembrane domain encoded by exon 11 (two-thirds of GIST). These mutations include in-frame deletions, substitutions, and insertions. Deletions (in particular codon 557 and/or 558) are associated with shorter progression-free and overall survival.^{1,2,3,4,5,6} The vast majority of exon 11-mutated GISTs occur in the stomach. About 7% to 10% of tumors harbor mutations in the extracellular domain encoded by exon 9 (most commonly insAY502-503).^{5,7} Exon 9-mutant GISTs arise predominantly in the small bowel and have reduced sensitivity to imatinib, which can be overcome by using higher doses.⁵ Primary mutations in the activation loop (exon 17) and ATP binding region (exon 13) are uncommon (1%). The majority of these mutations are substitutions.⁸ *KIT* exon 8 mutations are extremely rare (0.15%).⁹ Secondary (acquired) resistance mutations most commonly arise in tumors harboring primary exon 11 mutations. These mutations occur in exons encoding the *KIT* tyrosine kinase domain (exons 13, 14, 17, 18).¹⁰

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E. PDGFRA Mutational Analysis

More than 80% of *KIT*-negative GIST have *PDGFRA* mutations. The majority of *PDGFRA*-mutant GISTs arise in the stomach and demonstrate epithelioid or mixed epithelioid and spindle cell morphology, often with myxoid stromal changes.^{1,2} *PDGFRA*-mutant GISTs tend to have a lower recurrence risk.^{1,3} *PDGFRA* activation occurs through mutations in the juxta-membrane domain (exon 12), the ATP binding domain (exon 14), or the activation loop (exon 18).^{1,2} Mutations include substitutions and deletions. Primary resistance to imatinib is seen with *PDGFRA* exon 18 *D842V* mutation (the most common *PDGFRA* mutation in GIST).¹

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F. BRAF Mutational Analysis

Activating mutations of *BRAF* (V600E) have been identified in a small subset (7%) of *KIT*/*PDGFRA* wild-type GIST. These tumors show a predilection for the small intestine, arise in middle-aged females, exhibit a high mitotic rate, and are associated with early metastasis.^{1,2} *BRAF*-mutant GISTs show primary resistance to imatinib but may respond to *BRAF* inhibitors.²

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G. SDH A/B/C/D Mutational Analysis

The succinate dehydrogenase (SDH) complex (mitochondrial complex II) participates in both the Krebs cycle and the electron transport chain of oxidative phosphorylation. About 5% of GISTs (lacking mutations in *KIT* and *PDGFRA*) are caused by dysfunction of the SDH complex. Around 50% of patients affected by such tumors harbor germline mutations in one of the SDH subunit genes (*SDH A/B/C* or *D*). *SDHA*-inactivating mutations are most common, detected in about 30% of SDH-deficient GISTs. Mutations involve exons 2, 3, 5, 6, 7, 8, 9, 10, 11, 13, 14 of *SDHA*; exons 1, 2, 3, 4, 6, 7 of *SDHB*; exons 1, 4, 5 of *SDHC*; and exons 4 and 6 of *SDHD*. While the majority are substitutions, deletions, splice-site mutations, frameshift mutations, and duplications have also been reported.^{1,2,3,4}

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H. Neurofibromatosis Type 1 (NF1) Mutational Analysis

Neurofibromatosis type 1 is an autosomal dominant condition characterized by multiple café au lait spots, Lisch nodules, freckling, and development of multiple neurofibromas. 3 to 7% of NF1 patients develop GISTs, which arise in a background of Cajal cell hyperplasia, typically involve the duodenum, and are often multiple.^{1,2} Although NF1-associated GISTs nearly always lack *KIT* and *PDGFRA* mutations, molecular testing for *KIT* and *PDGFRA* is recommended when systemic therapy is contemplated.^{2,3,4,5}

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I. FGFR and BRAF Fusion Gene Analysis

Rare cases of GIST lacking the most common oncogenic driver events (i.e., *KIT*, *PDGFRA*, *SDHx*, *NF1* wild-type) have recently been reported to harbor fusions involving *FGFR1* or *BRAF*. Identification of these alterations may impact therapy.^{[1,2,3,4](#)}

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J. NTRK Fusion Gene Analysis

NTRK fusions in GIST are extremely uncommon. NGS or FISH testing is preferred to Pan-TRK immunohistochemistry for the detection of *NTRK* fusion in GIST.^{[1,2,3](#)}

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